

QSwitch: Reducing Interruption Time under WiFi-to-Cellular Transitions in P2P Conferencing

Kota Yatagai
Keio University
kota@sfc.wide.ad.jp

Kenshi Matsuya
Keio University
macchan@sfc.wide.ad.jp

Keisuke Uehara
Keio University
kei@sfc.wide.ad.jp

Abstract

Major conferencing services such as Zoom and Microsoft Teams use peer-to-peer (P2P) connection for one-to-one calls under certain conditions to reduce server-side media processing costs. However, P2P connectivity is fragile under an endpoint's WiFi-to-cellular transitions, where IP address changes often cause noticeable service interruptions. In this paper, we conduct a packet-level analysis of recovery procedures in existing P2P-based conferencing systems under IP address changes and find that repetitive transport-layer handshakes between an endpoint and a server dominate the interruption time. Motivated by these findings, we propose QSwitch, a seamless recovery approach under WiFi-to-cellular transitions for P2P conferencing. QSwitch leverages QUIC for both signaling and media delivery, taking advantage of its connection migration capability to enable an immediate fallback to a server-relayed path. Our evaluation shows that QSwitch reduces the median interruption time by 72% and 96% compared to Microsoft Teams and Zoom, respectively.

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1 Introduction

Real-time online conferencing has become essential for personal and professional communication [9, 18]. To reduce the growing infrastructure costs of large-scale media processing [5], many systems such as Zoom [1], Microsoft Teams [4],

Jitsi Meet [10] and Whereby [6] employ P2P connection for one-to-one calls under certain conditions¹².

However, P2P connectivity becomes fragile when a peer changes its network from WiFi to cellular. When a peer changes networks, its IP address changes and the existing NAT mapping becomes invalid. Restoring connectivity requires creating new mappings for the new address and exchanging updated addresses through a signaling channel, effectively repeating the establishment procedure. This leads to longer interruption time compared to server-centric systems, which can simply reconnect to global addresses of servers

To cope with the interruption under network transitions, several recovery approaches are proposed, including ICE Restart[13] or ICE Renomination[17]. Although these approaches differ in their mechanisms, they share a common limitation: media delivery remains blocked until signaling exchanges and transport-layer procedures complete.

To better understand the primary causes of the interruption, we analyze the recovery procedures and their overhead under IP address changes in P2P calls in Zoom and Microsoft Teams. We first investigate how each system recovers from an interruption by examining packet traces during a P2P call. We then decompose the recovery procedure into three phases: (i) the delay from the IP address change to recovery initiation, (ii) the delay until the remote peer becomes aware of the address change, and (iii) the delay until media delivery resumes. By measuring these delays, we find that the second phase dominates the overall interruption time across all services. Specifically, multiple transport-layer handshakes of TCP [8] and TURN [16] between multiple servers before signaling the peer contribute most to the overall interruption.

Motivated by these findings, we design QSwitch, a recovery approach for P2P conferencing under WiFi-to-cellular transitions. QSwitch leverages QUIC [11] to allow media delivery to continue across IP address changes without blocking on P2P re-establishment. When an IP address change occurs, QSwitch immediately restores connectivity via the

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¹Microsoft Teams enables P2P connection for one-to-one calls initiated via chat.

²Zoom enables P2P connection only when both participants are located within the same local network.

server by utilizing QUIC’s connection migration feature. Media transmission is then promptly migrated to the relay path, while QSwitch concurrently re-establishes a P2P path. Once the P2P path is restored, media traffic is switched back.

We implement an end-to-end prototype of QSwitch and evaluate it under WiFi-to-cellular transition scenarios during one-to-one calls. In our experiments, the median interruption time is 0.76s for QSwitch, compared to 2.68s for Microsoft Teams and 17.07s for Zoom. This demonstrates that QSwitch substantially reduces user-perceived interruption time after address changes.

In summary, this paper makes the following contributions:

- We conduct a packet-level analysis of existing P2P-based conferencing systems to characterize their behavior under IP address changes, and identify the key overhead of interruption.
- We propose QSwitch, a recovery approach from IP address changes that leverages QUIC to rapidly fall back to server-relayed media delivery.
- We evaluate QSwitch on mobile devices under realistic WiFi-to-cellular transition scenarios, and demonstrate that it substantially reduces the interruption time compared to existing conferencing services.

2 Background and Related Work

Many conferencing systems rely on ICE [13] to establish and maintain P2P connectivity. ICE gathers multiple candidate addresses, including local address, global address and relay address, exchanges them through a signaling server and performs connectivity checks to select a valid candidate pair for media transmission. Once a candidate pair is selected, media packets are sent over the chosen transport path.

Within ICE-based systems, two recovery approaches have been proposed. **ICE Restart** is the standard recovery approach defined in the ICE specification [13]. It regenerates ICE credentials and performs a new offer/answer exchange, effectively repeating the connectivity establishment process. Because it requires new signaling, candidate gathering, and connectivity verification, media delivery remains blocked until the entire procedure completes. **ICE Renomination** [17] reduces overhead existed in ICE Restart by reusing previously gathered candidate pairs without performing a full restart. However, it still requires connectivity checks and transport-layer handshakes between a signaling server before media transmission resumes.

Beyond conferencing systems, **server-relayed fallback** approach is used by several services including Tailscale [19]. In this approach, endpoints abandon P2P path upon address change and start packet transmission via an intermediate server. Since this approach does not seek to establish P2P

path after an IP address change, the recovery procedure can be simple and rapid.

As far as our knowledge, there is no prior work examining how deployed P2P conferencing systems actually recover from IP address changes at the packet level. In particular, the sources of interruption under WiFi-to-cellular transition remain insufficiently characterized. This motivates our empirical analysis of recovery procedures in existing systems at Section 3.

3 Characterizing Interruption Recovery

We first conduct a packet-level analysis of one-to-one calls in existing P2P-based conferencing systems to understand the recovery procedure after an IP address change and to identify the dominant sources of interruption. We focus on Zoom and Microsoft Teams and analyze their behavior under WiFi-to-cellular transition scenarios during ongoing calls. In each experiment, an Android device and a Macbook Pro join a one-to-one call, with the Mac continuously transmitting audio to the Android. During the call, WiFi is disabled on the Android device, forcing a transition to the cellular network and triggering an IP address change. The detailed experimental setup is described in Section 6.1.

To isolate the overhead, we decompose the recovery procedure into the following three phases:

- (1) **Detection delay:** the time from WiFi being disabled on the Android client to the initiation of the recovery procedure.
- (2) **Peer awareness delay:** the time from the initiation of the recovery procedure until the Mac client becomes aware of the Android’s IP address change.
- (3) **Media resumption delay:** the time from peer awareness until the first audio packet is received by the Android.

The timestamp of WiFi being disabled is taken from the `I/WifiService: enable=false` entry in the logcat output on the Android device. All other timing points are derived from packet traces by identifying protocol-specific events in the recovery procedure for each service. By correlating timestamps from packet traces with system logs, we compute the duration of each phase. A quantitative evaluation of interruption time and the phases are further presented in Section 6.

3.1 Zoom

Zoom enables P2P connection only when both participants are located within the same local network. Figure 1 shows the recovery sequence of a P2P call in Zoom under an IP address change. We observe that Zoom communicates with servers whose hostnames end with `mmr.nrt.zoom.us` (MMR) and `zpns.zoom.us` (ZPNS). The client establishes TLS 1.3 over

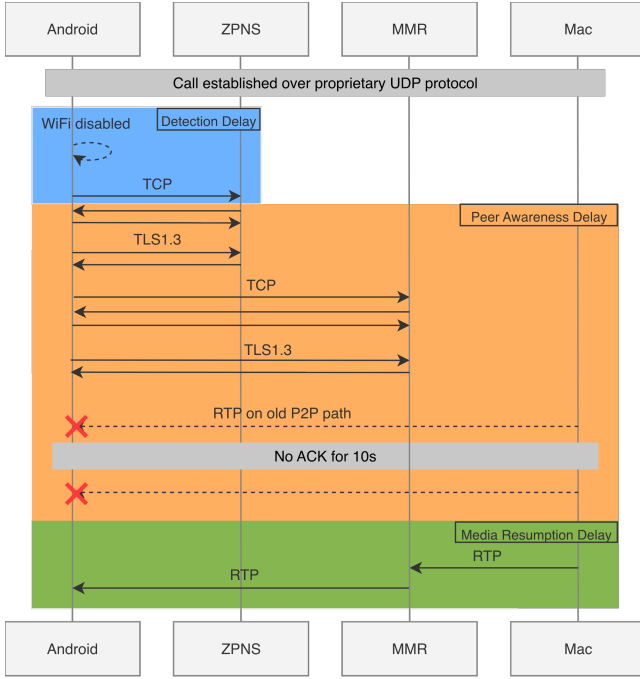


Figure 1: Zoom recovery sequence after an IP address change

TCP connections to both servers, and additionally communicates with MMR over UDP. From packet traces, MMR relays RTP packets both during session establishment and after reconnection, indicating that it functions as a Multi Media Router. Because TLS 1.3 payloads are encrypted, we cannot determine which server is responsible for signaling. During Zoom calls, we do not observe any STUN or TURN traffic, suggesting that Zoom employs a proprietary protocol instead of ICE, as described in other researches [12, 14]. Media transmission is observed over both the P2P path and the MMR-relayed path even after P2P path is established. On the P2P path, each transmitted message elicits a fixed 72-byte response packet, which we identify as an acknowledgment (ACK).

After WiFi is disabled, the Android reconnects to ZPNS using TLS 1.3 over TCP. We define the SYN packet toward ZPNS as the end of the *detection delay*. The Android then reconnects to MMR. The Mac continues transmitting over the P2P path for approximately 10s after the last ACK from the Android. Then it switches media transmission exclusively to the MMR-relayed path. We define the last packet sent over the P2P path as the end of the *peer awareness delay*. Although reconnections to ZPNS and MMR complete within 1s after WiFi is disabled, and RTP packets flow through MMR since the beginning, the Android does not start receiving RTP packets from MMR until the Mac stops P2P transmission.

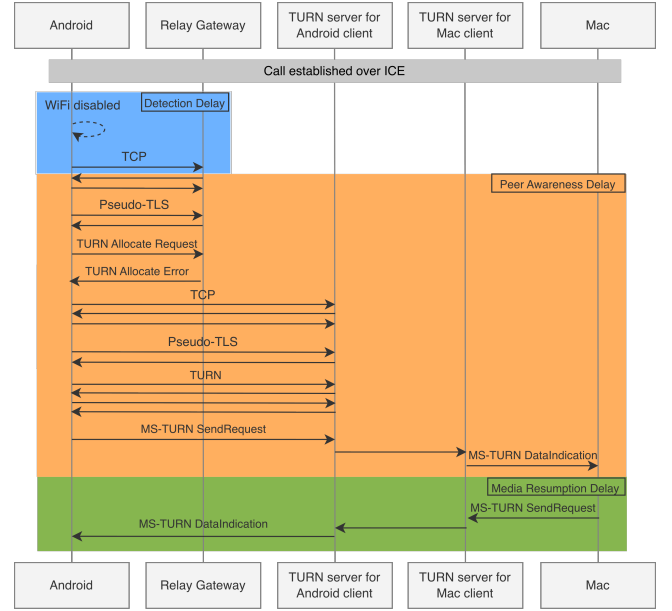


Figure 2: Microsoft Teams recovery sequence after an IP address change

The arrival of the first RTP packet via MMR at the Android marks the end of the *media resumption delay*.

As a result of the analysis, the *peer awareness delay* dominates the overall interruption time. This is because the Mac maintains the P2P path for approximately 10s after ACK loss and the Android does not receive audio packets from the server until then.

3.2 Microsoft Teams

Microsoft Teams enables P2P connection for one-to-one calls initiated via chat. Figure 2 shows the recovery sequence of a P2P call in Microsoft Teams under an IP address change. From packet traces, we observe that a Teams client communicates with world.relay.skype.com (Relay Gateway) and TURN servers associated with both clients using TCP and TURN. TCP handshakes to the servers are followed by a single round trip of pseudo-TLS handshake. While the clients also maintains connections with several other servers including a signaling server, these connections are not involved in the recovery sequence and are therefore omitted from the discussion.

Session establishment is performed using ICE, and our inspection of TURN message headers indicates that Teams does not comply with the standard defined in RFC8656 [16], but instead follows MS-TURN [2].

From NetworkRequest entry from dumphsys output, we observe that Microsoft Teams detects an IP address change from OS-level notification. After detecting an IP address

change, the Android first performs a TCP handshake with the Relay Gateway and sends a TURN Allocate Request. We define the SYN packet toward the Relay Gateway as the end of the *detection delay*. The Relay Gateway responds with an Allocate Error Response containing an ALTERNATE-SERVER attribute, redirecting the client to another TURN server. The client then sends a TURN Allocate Request to the indicated server and establishes a new TCP connection to it. In all trials, the redirected TURN server matches the relay candidate assigned to the Android prior to the IP address change. Almost simultaneously with the TURN reconnection, the Android sends STUN Binding Requests to previously gathered candidates without changing the username, indicating that Teams employs ICE Renomination.

On the Mac side, once the Android reconnects to the TURN server, a TURN Data Indication message is received. Because responses to STUN Binding Requests on the P2P path have already ceased, the relayed path becomes the only valid path, and the Mac immediately switches media transmission to the relay path and sends a Send Request. We define the reception of this Data Indication at the Mac as the end of the *peer awareness delay*. After the Mac switches media transmission, a corresponding Data Indication carrying media arrives at the Android, which we define as the end of the *media resumption delay*.

The analysis shows that the *peer awareness delay* dominates the overall interruption time. This overhead results from repeated TCP and TURN connection establishments toward the Relay Gateway and the redirected TURN server.

3.3 Key Insights

Zoom exhibits the longest interruption because the sender peer waits for 10s to give up on the P2P path. Teams reduces interruption by employing ICE Renomination. However, Teams still suffers from multiple repetitive handshakes of TCP and TURN between servers. This suggests that reducing interruption requires avoiding repeated server connections and transport re-establishment during recovery.

4 QSwitch Design

QSwitch is a QUIC-based recovery approach under WiFi-to-cellular transitions for P2P conferencing systems. Because QUIC [11] manages connections using Connection IDs rather than the conventional 4-tuple, a client can resume an existing connection after an IP address change with a single round-trip path validation. Leveraging the insight from Section 3 that the transport-layer handshakes dominate interruption time, QSwitch uses QUIC connection migration to immediately reestablish connectivity with the server upon an address change, notify the remote peer, and fall back to server-relayed media transmission. This design minimizes

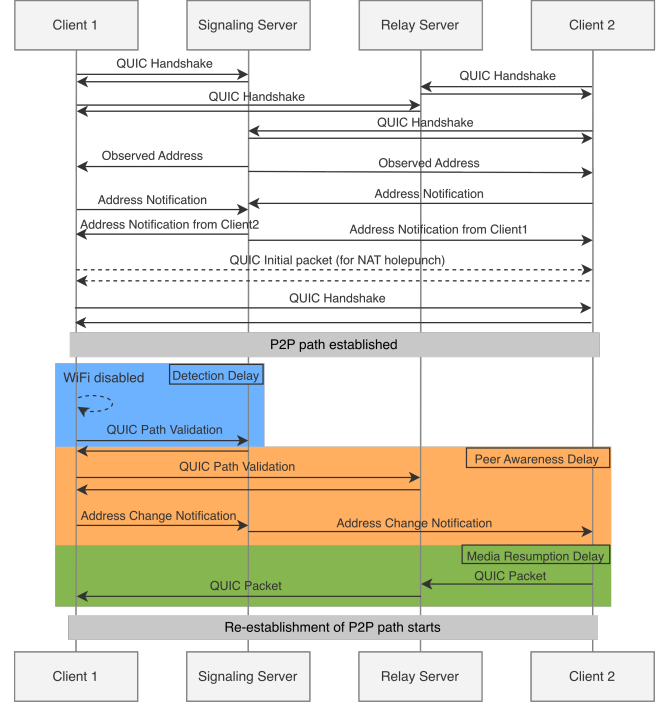


Figure 3: Sequence of QSwitch connection establishment and recovery procedure

the *peer awareness delay*, the delay before the remote peer becomes aware of the address change.

4.1 P2P Session Establishment

The upper half of Figure 3 illustrates the P2P session establishment sequence of QSwitch. Each client first establishes QUIC connections to both the signaling server and the relay server. The signaling server returns the externally observed address of the client. Each client then sends an Address Notification, containing its local and global addresses, to the remote peer via the signaling server.

Upon receiving the peer's addresses, each client sends QUIC Initial packets to both the local and global addresses of the peer. Connectivity attempts prioritize the local address over the global address. Once a valid path is established, media transmission proceeds over the P2P path.

4.2 Recovery Procedure

The lower half of Figure 3 illustrates the recovery sequence under an IP address change. When a client's IP address changes, it detects the event immediately via a netlink notification. The client then resumes its QUIC connections with the signaling server and the relay server using a single round-trip path validation enabled by QUIC connection migration.

After reconnecting to the signaling server, the client sends an Address Change Notification, including its updated local address, to the remote peer via the signaling server. The signaling server appends the newly observed global address and forwards the notification to the remote peer. Upon receiving the notification, the remote peer immediately switches media transmission to the relay server. Media delivery is thus restored via the relay path.

After media transmission resumes, the peers re-establish the P2P path using the same procedure described in Section 4.1. Once a valid P2P path is restored, media transmission is switched back from the relay path.

5 Implementation

QSwitch is an end-to-end recovery approach that requires coordinated modifications to both clients and servers. Since it fundamentally redesigns signaling, relaying, and connection establishment during a call, it cannot be incrementally integrated into existing open-source conferencing systems such as Jitsi Meet[3]. Therefore, we implement QSwitch as a standalone prototype. The client and servers are written in Go and built on top of quic-go (v0.58.0).

A QSwitch client consists of approximately 2,500 lines of code and runs on both Linux and macOS. It is composed of several components coordinated by an orchestrator module that manages connection state and peer address information. A monitoring component detects IP address changes using netlink notifications. Upon detecting an address change, it notifies the orchestrator, which triggers the recovery procedure described in Section 4.2.

The orchestrator manages three main components. Hole punch component is responsible for NAT hole punching to establish and re-establish the P2P path. Control plane component handles signaling messages exchanged with the signaling server, including Address Notification and Address Change Notification. Data plane component manages media transmission over QUIC connections, including switching between the P2P path and the relay path. The prototype supports audio transmission from an MP3 file.

QSwitch uses two server components: a signaling server and a relay server. Both consist of fewer than 400 lines of code each. The signaling server maintains QUIC connections with clients and forwards control messages between peers. The relay server forwards media packets between peers when P2P path becomes invalid.

6 Evaluation

We evaluate the effectiveness of QSwitch in reducing interruption time under IP address changes. We first measure

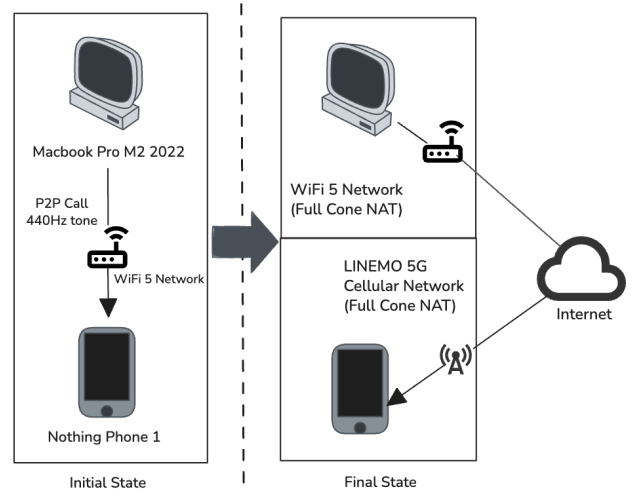


Figure 4: Experimental setup for recovery analysis.

user-perceived interruption time by measuring silence duration from audio recordings. Second, we analyze its breakdown across phases to understand where improvements are achieved.

6.1 Measurement Setup

Figure 4 illustrates our experimental setup. To ensure a consistent comparison across services, both endpoints are placed on the same WiFi 5 local network, since Zoom enables P2P connection only when participants share a local network. The WiFi network is configured with a full-cone NAT, and the LINEMO cellular network is also observed to employ a full-cone NAT.

The experiments involve a Nothing Phone 1 running Android 13 and a MacBook Pro (M2, 2022) running macOS Tahoe 26.2. During each trial, the Mac continuously transmits a 440 Hz tone to the Android device. To emulate network transition, WiFi is disabled on the Android during an ongoing call, forcing a transition to the cellular network and triggering an IP address change. Packet traces are collected on both endpoints, and logcat logs on the Android device are used to determine the exact timing of WiFi disablement. Prior to each experiment, we synchronize the clocks of both devices using the Google NTP server at `time.google.com` to ensure consistent timestamp correlation across packet traces and system logs. We also confirm that the round-trip times to all servers contacted by QSwitch, Microsoft Teams, and Zoom are below 20ms, ensuring that network latency does not significantly influence the measured interruption time.

Table 1: User-perceived interruption time

Service	Median (s)	Max (s)	Min (s)
QSwitch	0.762	1.069	0.594
Teams	2.682	2.851	2.142
Zoom	17.072	18.539	10.101

6.2 User-perceived Interruption Time

To quantify interruption time from the user’s perspective, we record the audio received at the Android device and measure periods of silence in each recording. Silence is detected automatically by treating samples below -35dBFS as inaudible. The longest silent interval in a recording is defined as the interruption time for that trial.

Table 1 shows the median user-perceived interruption time over 10 runs for each system. QSwitch achieves 0.762s, whereas Microsoft Teams and Zoom exhibit 2.682s and 17.072s, respectively. These results indicate that QSwitch substantially reduces user-perceived interruption time following a peer’s IP address change.

6.3 Breakdown of Interruption by Phase

To further analyze the sources of interruption, we decompose the interruption into the three phases defined in Section 3. For Zoom and Microsoft Teams, the phase boundaries are determined using the event definitions identified in Section 3. For QSwitch, each phase boundary is defined as follows:

- **Detection delay:** the time from WiFi being disabled on the Android device to the transmission of a QUIC path validation packet toward the intermediate server.
- **Peer awareness delay:** the time from the transmission of the path validation packet to the reception of the Address Change Notification at the Mac.
- **Media resumption delay:** the time from the reception of the notification at the Mac to the reception of media packets at the Android device.

Figure 5 presents the breakdown of interruption time across the three phases for each system. Each value represents the median duration over 10 runs for each service. For the detection delay, all systems achieve delays of less than one second due to rapid OS-level notifications triggered by IP address changes. For the peer awareness delay, QSwitch achieves 0.177s by immediately validating a new path using QUIC connection migration, and signaling the remote peer. Microsoft Teams exhibits a delay of 1.611s due to the need to establish new TCP and TURN sessions with multiple relay servers. Zoom incurs a delay of 10.168s, primarily because the Mac continues attempting to use the outdated P2P path for approximately 10s after losing acknowledgments before switching to the relay path. For the media resumption delay,

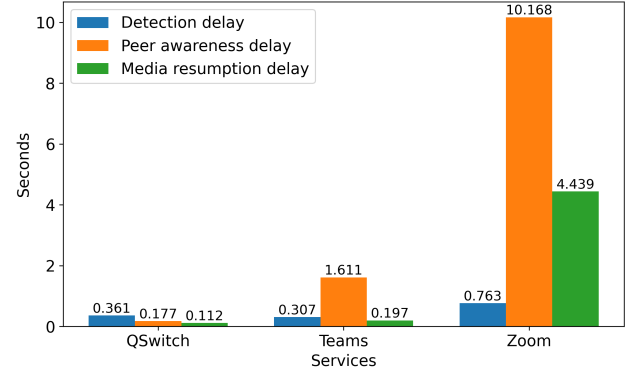


Figure 5: Breakdown of interruption time by phase. Each value is the median of 10 trials.

QSwitch and Microsoft Teams exhibit similar delays of 0.112s and 0.197s, respectively. Zoom shows a significantly longer delay of 4.439s because the Android does not open a media channel to the server until signaling indicates that the P2P path has been abandoned.

7 Limitation

Emulated WiFi-to-Cellular Transitions. In our experiments, network transition is emulated by explicitly disabling the WiFi interface, which causes an immediate switch to the cellular network. In real deployments, however, connectivity typically degrades gradually due to worsening signal quality or increasing packet loss, and the operating system switches the primary interface only after sustained degradation rather than a discrete failure. Consequently, interruption behavior in practice may differ from that observed in our controlled setup. Future work should incorporate mechanisms for continuously monitoring network conditions and proactively initiating path migration.

Media Transport over QUIC. Another limitation of QSwitch is the use of QUIC for media transport. Modern real-time conferencing systems are largely built on WebRTC[7], whose major implementations provide mature and highly optimized media stacks including RTP-based media transport, congestion control, jitter buffering, packet pacing, and adaptive bitrate mechanisms integrated with ICE-based connectivity establishment. These components have been refined over 10 years of deployment.

In contrast, QUIC primarily provides a general-purpose transport protocol and does not natively include media-layer mechanisms comparable to those available in WebRTC. While Media over QUIC[15] is proposed as a payload-agnostic media transport protocol over QUIC, deploying QUIC-based media transport in production conferencing systems would

still require substantial engineering effort to match the performance and robustness of existing WebRTC-based solutions.

8 Conclusion

We analyze recovery procedures in existing P2P conferencing systems under IP address changes and identify that the interruption time is dominated by signaling and transport-layer procedures, particularly the delay before the remote peer becomes aware of the address change. Based on this insight, we present QSwitch, a QUIC-based mechanism that enables rapid fallback without repeated transport handshakes. Evaluations on real devices demonstrate that QSwitch substantially reduces user-perceived interruption compared to existing systems.

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